

Politecnico di Milano
Biochip A.Y. 2020/2021 - M. Carminati - Exam of 16.06.2021

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Consider the following microfluidic device (Fig. 1) for the study of electrical properties of mammalian cells. It consists of a bifurcating channel on top of two sets of planar electrodes (of the same size): a first couple of *dynamic* electrode to detect in flow by impedance the passage of a cell and static ones to measure its impedance evolution once it is trapped in a constriction (C). Once a cell is trapped, liquid flows in the bypass E-F. The channel height is h everywhere. The channel width is w in all sections, except for the constriction C where it is w_c .

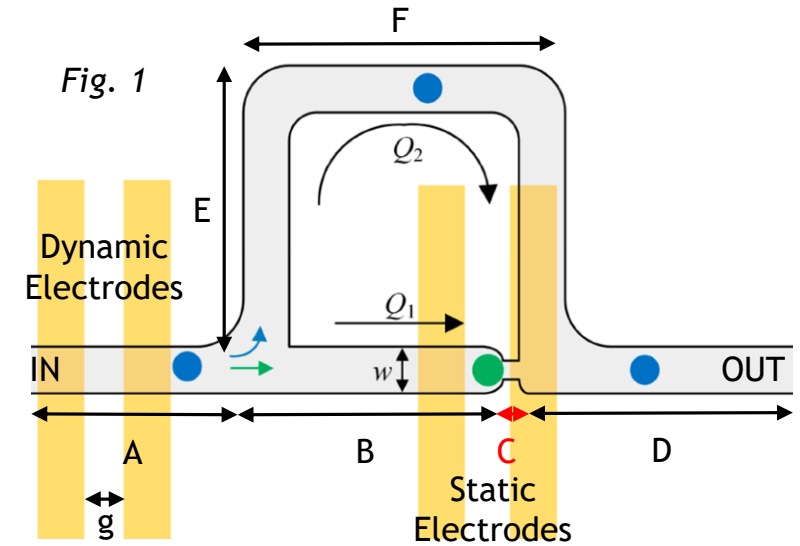
Part 1

- Find (draw and compute the value of the parameters) the electrical equivalent model of the device with both no cell and cell trapped.
- Assuming to cascade several identical devices, what is the max. number of devices compatible with a total pressure drop of 5 atm and a flow rate of 4 $\mu\text{L}/\text{min}$.
- From Fig. 2, estimate the value of the input flow rate in this case.
- In order to de-trap the cell, a flow could be applied from the OUT. Is the fluidic resistance the same when pushing from the IN port and from the OUT port?
- How does the model change if the cells are in blood instead of water and if an air bubble gets trapped (clogging C) instead of a cell?
- Describe the steps for the fabrication of the device.

Data:

A = 1 mm
 B = 2 mm
 C = 20 μm
 D = 1 mm
 E = 2 mm
 F = 2 mm
 g = 10 μm
 w = 50 μm
 h = 10 μm
 $w_c = 2 \mu\text{m}$

1 atm = 101 kPa
 $\eta_{\text{water}} = 10^{-3} \text{ Pa}\cdot\text{s}$



Part 2

- Assuming a PBS solution and gold electrodes, find the small-signal equivalent impedance model between the electrodes (with no cell).
- What is the impedance model of the cell and how does it affect the measured spectrum?
- From the measured impedance spectrum of Fig. 3, extract the values of the parameters of model of question (g). What is the length of the electrodes? What is the optimal measuring frequency for counting passing cells at *dynamic* electrodes?
- Draw the scheme of the impedance readout circuit and choose the values of the components (assume $V_{DD} = -V_{SS} = 3 \text{ V}$).
- Assuming an input flow rate of 1 $\mu\text{L}/\text{min}$, find the bandwidth of the LPF in the lock-in amplifier connected the *dynamic* electrodes.
- Considering only the noise of the feedback resistor of the TIA, find its value to achieve an input resolution of 100 pA_{rms}.

